Senior Battery Data Engineer

Redwood Materials · San Francisco, California, United States

Job ID (this archive)	ENERGY-100
Employer	Redwood Materials
Location(s)	San Francisco, California, United States
Career stage (archive tag)	Experienced (Senior)
Location group	Other U.S.
Majors (archive tags)	Electrical Engineering, Mechanical Engineering, Data Science
Energy area	Battery recycling and storage / data
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/redwoodmaterials/jobs/6219757004?gh_jid=6219757004
Original link (from the course sheet)	https://boards.greenhouse.io/redwoodmaterials/jobs/6219757004?gh_jid=6219757004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-071

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Redwood Materials

Redwood Materials was founded in 2017 by JB Straubel, co-founder and former CTO of Tesla, on a simple but radical idea: the critical minerals powering the energy transition already exist. They just need to come back. Today Redwood is the largest lithium-ion battery recycler in North America — recovering, refining, and redeploying critical minerals from end-of-life batteries back into the supply chain. On the energy side, Redwood has deployed the largest second-life battery project in the world, delivering power to AI data centers in months rather than years using proprietary energy architecture engineered entirely in-house. The company is profitable, the market it operates in is projected to grow 20 to 50 times over the next decade, and Redwood is the only company with the supply chain, the technology, and the operational scale to lead it.

Position Summary:

The Senior Battery Data Engineer develops the software stack that powers Redwood's battery characterization program, spanning data pipelines, hardware control services, and workflows that connect test data to downstream modeling and analysis. This is a fully on-site role at 100 Hooper. As the software stack matures, this role has the opportunity to shape it into a deterministic tool-calling layer, leveraging it to build an AI agent co-pilot that accelerates the characterization workflow, with evals co-developed alongside battery domain experts. This role sits at the intersection of data infrastructure and systems automation, building the foundational software that powers battery modeling, algorithm, and qualification efforts.

Essential Duties and Responsibilities

- Own the data infrastructure that serves large-scale battery time-series data from battery cyclers to cloud storage
- Contribute to the software services that orchestrate testing using commercial lab battery cyclers and in-house field-level power converters located at production sites
- Develop and maintain the tool layer for battery testing, such as structured protocol schemas, cycler control APIs, real-time battery test anomaly detection, and digital twin simulation
- Partner with battery test and algorithm engineers to identify inefficiencies, then design and build software or agentic workflows to automate them, utilizing the deterministic tool layer above

- Collaborate with domain experts to define agent eval suites and correctness criteria as agentic automation is introduced

Skills, Knowledge, and Abilities

- BS or MS in Engineering (Mechanical, Chemical, Materials, Electrical), or a related field
- Strong software development skills in Python, including building production services, data pipelines, and APIs
- Familiarity with battery testing and cycler platforms (e.g., Arbin, Maccor, Neware)
- Experience with cloud data infrastructure (e.g., AWS, GCP, or Azure) for storing, processing, and serving large-scale time-series data
- Strong communication and collaboration skills; proactive in working directly with stakeholders to identify workflow bottlenecks and turn them into actionable software priorities
- Bonus: proficiency in Rust for building production services; experience building AI agent harnesses and eval suites; understanding of lithium-ion cell behavior and safety considerations; experience with AI coding assistants like Cursor or Claude Code

In accordance with California pay transparency laws, the salary range for this position is listed below. Actual compensation may vary based on a variety of factors, including experience, education, and skills.

California Pay Range:

$172,500—$198,000 USD

The position is full-time. Compensation will be commensurate with experience.

We collect personal information (PI) from you in connection with your application for employment with Redwood Materials, including the following categories of PI: identifiers, personal records, professional or employment information, and inferences drawn from your PI. We collect your PI for our purposes, including performing services and operations related to your potential employment. If you have additional privacy-related questions, please contact us at privacy@redwoodmaterials.com.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: BS or MS in Engineering (Mechanical, Chemical, Materials, Electrical), or a related field

Key skills listed in the archive: Python; Rust; AWS; Azure; time-series; automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6219757004

Career stage: Senior

Major fit: Direct skills pathway: data science, machine learning, analytics or data engineering; exact degree/experience requirements retained. Energy-sector connection: Computer vision for battery sorting or data engineering for energy-storage batteries.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6219757004.

Sources and retrieval details

Employer posting

https://boards.greenhouse.io/redwoodmaterials/jobs/6219757004?gh_jid=6219757004

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https://boards-api.greenhouse.io/v1/boards/redwoodmaterials/jobs?content=true

Retrieved: 2026-10-26T16:33:09.192514+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.